

HORIZON MEDICAL — Clinical Evaluation Report (CER)

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1. Scope of the Clinical Evaluation

1.1 Purpose

This Clinical Evaluation Report (CER) evaluates the clinical safety and performance of the Horizon Medical Holter ECG System (HM-H100) in accordance with the requirements of the Medical Device Regulation (EU) 2017/745, Article 61, Annex XIV Part A, and MEDDEV 2.7/1 Revision 4.

1.2 Device Information

Parameter	Description
Device name	Horizon Medical Holter ECG System
Model	HM-H100 v1.0
Classification	Class IIb (MDR Rule 11)
Intended purpose	Ambulatory ECG monitoring (24-48h) with AI-assisted arrhythmia detection
Target population	Adult patients (≥ 18 years) with suspected or known cardiac arrhythmias
Clinical evaluator	[Name, MD, FACC — qualifications and CV in Appendix A]

1.3 Scope of Evaluation

This CER covers:

- Clinical safety of the Horizon Medical Holter ECG System during ambulatory use
- Clinical performance of ECG signal acquisition and quality
- Performance of the AI/ML module for arrhythmia detection
- Benefit-risk profile for the intended population and use

2. Clinical Background and Current Knowledge

2.1 Cardiac Arrhythmias — Epidemiology

Cardiac arrhythmias are a leading cause of morbidity and mortality worldwide. Atrial fibrillation (AF) alone affects approximately 33.5 million people globally, with prevalence increasing with age. Ventricular tachycardia (VT) and ventricular fibrillation (VF) are life-threatening arrhythmias that require immediate medical attention.

2.2 Ambulatory ECG Monitoring — State of the Art

Ambulatory ECG monitoring (Holter monitoring) has been the gold standard for arrhythmia detection since its introduction by Norman Holter in 1961. Modern Holter systems provide:

- Continuous ECG recording for 24-48 hours (or longer with patch-type devices)
- Digital signal processing with computerized analysis
- Arrhythmia detection algorithms with varying degrees of accuracy
- Post-hoc analysis by trained technicians and cardiologists

2.3 AI/ML in Cardiac Monitoring

The use of artificial intelligence and machine learning algorithms for ECG analysis has been extensively studied. Deep learning approaches (CNN, LSTM, and hybrid architectures) have demonstrated

performance comparable to or exceeding that of human experts for specific arrhythmia detection tasks.

Key published evidence:

Study	Algorithm	Performance	Reference
Hannun et al. (2019)	DNN (34-layer CNN)	Cardiologist-level AF detection (F1: 0.837)	Nature Medicine 25(1):65-69
Ribeiro et al. (2020)	DNN	6 arrhythmia classes, AUC >0.95	Nature Communications 11:1760
Attia et al. (2019)	CNN	AF detection from sinus rhythm ECG, AUC 0.87	The Lancet 394(10201):861-867
Yildirim et al. (2018)	1D-CNN	17 arrhythmia classes, accuracy 91.3%	Computers in Biology and Medicine 102:411-420

2.4 Equivalent and Predicate Devices

Multiple ambulatory ECG monitoring systems with computerized arrhythmia detection are currently on the market with established clinical evidence:

Device	Manufacturer	Regulatory Status	Clinical Evidence
Zio XT / Zio AT	iRhythm	FDA cleared, CE marked	Multiple published studies, >300,000 patients
MCOT	BioTelemetry/Philips	FDA cleared, CE marked	Extensive clinical use
Cardea SOLO	Cardiac Insight	FDA cleared	Published validation studies
KardiaMobile 6L	AliveCor	FDA cleared, CE marked	Multiple published studies

3. Device Under Evaluation

3.1 Device Description

The Horizon Medical Holter ECG System consists of:

1. **Wearable device (HM-H100):** Portable ECG acquisition unit with 3-channel Ag/AgCl surface electrodes, nRF52832 SoC, ADS1298 AFE, BLE connectivity, LiPo battery (48h+ autonomy)

- 2. **Mobile application:** Patient-facing app (iOS/Android) for real-time ECG visualization, symptom logging, and alert notifications
- 3. **Cloud backend:** Data processing, storage, and AI/ML analysis engine
- 4. **AI module (CNN-LSTM):** Automated arrhythmia detection and classification algorithm
- 5. **Clinical dashboard:** Web-based interface for healthcare professionals to review ECG data, AI results, and generate clinical reports

3.2 Intended Purpose

The system is intended for ambulatory ECG monitoring of adult patients for 24–48 hours, with AI-assisted detection of cardiac arrhythmias as a clinical decision support tool. All AI results must be reviewed and confirmed by a qualified healthcare professional.

3.3 Device Generations

Version	Date	Changes
HM-H100 v1.0	2026	Initial market release

4. Clinical Evaluation Plan

4.1 Literature Search Strategy

Parameter	Detail
Databases	PubMed/MEDLINE, Cochrane Library, Embase, IEEE Xplore
Search period	2010 — 2026
Languages	English, Spanish
Search terms	(“Holter monitor” OR “ambulatory ECG” OR “continuous ECG monitoring”) AND (“ar-rhythmia detection” OR “atrial fibrillation” OR “ventricular tachycardia”) AND (“artificial intelligence” OR “machine learning” OR “deep learning” OR “CNN” OR “LSTM”)
Inclusion criteria	Clinical studies, systematic reviews, meta-analyses involving ambulatory ECG with AI/ML analysis in adult patients
Exclusion criteria	Case reports with <10 patients, non-peer reviewed, conference abstracts only, pediatric populations

4.2 Data Sources

Source	Type	Purpose
Published literature	External	Safety and performance of equivalent devices
Bench testing data	Internal	Device performance verification
AI validation data	Internal	Algorithm performance metrics
Usability study data	Internal	User acceptance and error analysis
Clinical validation study	Internal	Head-to-head comparison with standard Holter

4.3 Appraisal Criteria

Each identified article was appraised using the following criteria:

- **Suitability:** Relevance to the device under evaluation and its intended purpose
- **Contribution:** Quality of data provided (study design, sample size, statistical methods)
- **Scientific validity:** Methodological rigor, bias assessment, reporting quality

5. Clinical Data from Literature

5.1 Literature Search Results

Step	Results
Initial search results	[N] articles identified
After duplicate removal	[N] articles
After title/abstract screening	[N] articles
After full-text review	[N] articles included
Additional hand-search	[N] articles
Total included	[N] articles

5.2 Summary of Key Findings — Safety

Finding	Evidence	Strength
Ambulatory ECG monitoring with surface electrodes is safe	Multiple large studies (>100,000 patients), no serious adverse events related to the device	Strong
Skin irritation from adhesive electrodes occurs in 5–15% of patients	Multiple studies; self-limiting, mild severity	Moderate
No electrical safety events (shock, burns) reported with modern Holter devices	Regulatory databases (MAUDE, vigilance), literature	Strong
False positive alerts may cause transient anxiety	Reported in studies of AI-based monitoring	Moderate

5.3 Summary of Key Findings — Performance

Finding	Evidence	Strength
Modern Holter ECG provides diagnostic-quality 3-channel ECG recordings	Established technology, extensive clinical use	Strong
AI/ML algorithms achieve sensitivity $\geq 90\%$ for AF detection	Multiple published studies (Hannun 2019, Ribeiro 2020)	Strong
AI/ML algorithms achieve sensitivity $\geq 90\%$ for VT detection	Published studies (Ribeiro 2020, Yildirim 2018)	Moderate
CNN-LSTM hybrid architectures show superior performance to traditional algorithms	Comparative studies in literature	Moderate
Patch/wearable ECG provides comparable data quality to traditional Holter	Published comparison studies	Moderate

6. Clinical Data from the Device

6.1 Bench Testing Data

Test	Standard	Result	Conclusion
ECG signal accuracy	IEC 60601-2-47	[Results]	Meets requirements
Frequency response	IEC 60601-2-47	[Results]	Meets requirements
CMRR	IEC 60601-2-47	≥ 100 dB	Meets requirements
Noise level	IEC 60601-2-47	≤ 30 μ V pp	Meets requirements
Battery endurance	Internal spec	≥ 48 hours	Meets requirements

6.2 AI Algorithm Validation Data

Metric	AF	VT	Bradycardia	Tachycardia	PVC
Sensitivity	[%]	[%]	[%]	[%]	[%]
Specificity	[%]	[%]	[%]	[%]	[%]
PPV	[%]	[%]	[%]	[%]	[%]
NPV	[%]	[%]	[%]	[%]	[%]
AUC-ROC	[value]	[value]	[value]	[value]	[value]
F1 Score	[value]	[value]	[value]	[value]	[value]
Validation dataset	[N records]	[N records]	[N records]	[N records]	[N records]

Note: Validation performed on independent dataset not used in training. Dataset sources: [PhysioNet MIT-BIH, INCART, proprietary clinical dataset — details in AI/ML Documentation]

6.3 Clinical Validation Study Data

Parameter	Result
Study design	Prospective, comparative, blinded interpretation
Comparator	Standard 12-lead Holter ECG
Sample size	N = [number] patients
Demographics	Age: [mean $\pm$ SD], Sex: [M/F ratio], BMI: [mean $\pm$ SD]
Monitoring duration	[mean $\pm$ SD] hours
Diagnostic concordance	[%] (κ = [value])
Adverse events	[number and description]

6.4 Usability Study Data

Parameter	Result
Participants — patients	N = [number]
Participants — clinicians	N = [number]
Task completion rate	[%]
Use errors (critical)	[number]
Use errors (non-critical)	[number]
Effectiveness	[assessment]

7. Equivalence Assessment

7.1 Equivalence with iRhythm Zio XT

Criterion	Horizon HM-H100	Zio XT	Equivalent?
Clinical	Same intended purpose (ambulatory ECG monitoring + arrhythmia detection)	Ambulatory ECG + arrhythmia detection	✓ Yes
Clinical	Same target population (adults with suspected arrhythmias)	Adults with suspected arrhythmias	✓ Yes
Clinical	Same anatomical site (chest)	Chest (sternal)	✓ Yes
Technical	3-channel ECG, Ag/AgCl electrodes	1-channel ECG, integrated electrodes	⚠ Similar (more channels)
Technical	BLE real-time + cloud processing	Store-and-forward	⚠ Different (real-time is additional feature)
Technical	AI (CNN-LSTM) analysis	Proprietary algorithm	⚠ Similar (different algorithm)
Biological	Ag/AgCl electrodes, ABS/PC housing	Ag/AgCl electrodes, medical adhesive	✓ Similar

7.2 Equivalence Conclusion

The Horizon Medical HM-H100 demonstrates **clinical equivalence** with the Zio XT and other predicate devices. Technical differences (additional channels, real-time connectivity, AI architecture) represent **improvements** that do not raise new safety concerns and are supported by the bench testing and AI validation data presented in Section 6.

Note regarding MDR: Under MDR, full equivalence demonstration requires contractual access to the equivalent device's clinical data if the manufacturer is different. Since Horizon Medical and iRhythm are different entities, this CER relies primarily on published literature and the device's own clinical data to support safety and performance claims, supplemented by the equivalence assessment.

8. Analysis of Clinical Safety

8.1 Identified Risks and Adverse Events

Risk/Event	Frequency (Literature)	Severity	Mitigation
Skin irritation from electrodes	5-15%	Mild, self-limiting	Hypoallergenic adhesive, IFU instructions
False positive alarm (anxiety)	Variable (dependent on threshold)	Mild	Physician confirmation required
False negative (missed arrhythmia)	1-5% depending on algorithm	Potentially serious	High Se algorithm, physician review
Signal artifact (motion)	Common during activity	N/A (data quality)	Artifact detection + filtering
Device detachment	1-5%	None (data loss only)	Impedance monitoring + alert

8.2 Safety Conclusion

Based on the literature review, bench testing data, and clinical validation results, the Horizon Medical Holter ECG System demonstrates an **acceptable safety profile** consistent with currently marketed ambulatory ECG monitoring devices. No new or unexpected safety concerns have been identified.

9. Analysis of Clinical Performance

9.1 ECG Signal Quality

The device demonstrates diagnostic-quality ECG signal acquisition as verified by bench testing according to IEC 60601-2-47.

9.2 AI Arrhythmia Detection Performance

Arrhythmia	Target	Achieved	Meets Target?
AF sensitivity	≥ 95%	[%]	☐ Yes / ☐ No
AF specificity	≥ 95%	[%]	☐ Yes / ☐ No
VT sensitivity	≥ 95%	[%]	☐ Yes / ☐ No
VT specificity	≥ 98%	[%]	☐ Yes / ☐ No
Overall AUC	≥ 0.95	[value]	☐ Yes / ☐ No

9.3 Performance Conclusion

The Horizon Medical Holter ECG System demonstrates clinical performance that is [equivalent to / superior to] currently marketed Holter ECG systems for the intended purpose of ambulatory ECG monitoring and arrhythmia detection.

10. Benefit-Risk Analysis

10.1 Benefits

1. **Detection of clinically significant arrhythmias** during ambulatory monitoring
2. **AI-assisted analysis** reducing time to detection and potential for human error
3. **Real-time alerts** for critical arrhythmias enabling faster clinical intervention
4. **Remote monitoring** reducing need for in-person visits
5. **Patient comfort** with compact, lightweight design improving adherence

10.2 Risks

1. **False negative** detection (mitigated by high sensitivity algorithm + physician review)
2. **False positive** alerts (mitigated by physician confirmation)
3. **Skin irritation** from electrodes (mitigated by hypoallergenic materials)
4. **Data privacy** concerns (mitigated by encryption and access controls)

10.3 Benefit-Risk Conclusion

The benefits of the Horizon Medical Holter ECG System **clearly outweigh** the identified residual risks for the intended population and use. The benefit-risk profile is consistent with or superior to currently marketed equivalent devices.

11. Conclusions

11.1 Summary

This Clinical Evaluation Report has assessed the clinical safety and performance of the Horizon Medical Holter ECG System (HM-H100) based on:

- Systematic literature review of [N] publications
- Bench testing data demonstrating compliance with IEC 60601-2-47
- AI algorithm validation data demonstrating high sensitivity and specificity
- Clinical validation study data (N = [number] patients)
- Equivalence assessment with marketed Holter ECG devices
- Comprehensive risk management (ISO 14971)

11.2 Conclusions

1. The Horizon Medical Holter ECG System is **safe** for its intended purpose of ambulatory ECG monitoring in adult patients
2. The AI module demonstrates **clinically acceptable performance** for arrhythmia detection as a decision support tool
3. The **benefit-risk profile is favorable** and consistent with the state of the art

4. The device meets the **General Safety and Performance Requirements** of MDR 2017/745 Annex I

11.3 Recommendations

- 1. Update this CER at least annually or upon significant changes to the device
- 2. Implement the PMCF Plan to collect ongoing clinical data
- 3. Monitor post-market data for emerging safety signals
- 4. Consider additional clinical studies to strengthen evidence for AI performance in specific subpopulations

11.4 Clinical Evaluator

Field	Detail
Name	[Name]
Qualifications	[MD, FACC, Board Certified Cardiologist]
Experience	[Years of experience in cardiac electrophysiology and device evaluation]
Affiliation	[Institution]
Date	[Date]
Signature	_____

Control de Versiones:

Versión	Fecha	Autor	Descripción
1.0	2026-05-16	[Clinical Evaluator]	Initial CER